

A COMMENT ON TOPOLOGICAL FIELD THEORY QUANTIZATION AND STOCHASTIC PROCESSES

L. F. CUGLIANDOLO, G. LOZANO, H. MONTANI* and F. A. SCHAPOSNIK†

Depto. de Física — Universidad Nacional de La Plata, C.C.67, 1900 La Plata, Argentina

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We discuss the relation between different quantization approaches to topological field theories by deriving a connection between Bogomol'nyi and Langevin equations for stochastic processes which evolve towards an equilibrium state governed by the topological charge.

1. Introduction

The connection between topological quantum field theories (TQFTs) and the instanton defining equation was originally stressed by Witten.¹ One way of understanding this connection is to consider the instanton equation as a gauge fixing condition in the BRST quantization of a topological action (invariant under the diffeomorphism group, thus requiring a gauge fixing). This is the approach advocated by Baulieu and Singer² as a way of arriving at TQFT Lagrangians.

Another point of view is that TQFTs can be obtained by BRST quantization of a trivial Gaussian action where instanton equations (Bogomol'nyi equations, in the general case) come into play.^{3,4} The authors in Ref. 4 identified this Gaussian action with the square of some Langevin equation; this observation allowed the construction of Nicolai maps for several interesting models.⁴ The relation between TQFTs and the stochastic quantization was also analyzed in Ref. 5. Alternative approaches to TQFTs were presented in Refs. 6–8.

In the present work we discuss the connections between these different viewpoints by deriving a relation between Langevin and Bogomol'nyi equations. From this, the link between Baulieu-Singer² and Labastida-Pernici³ approaches to TQFTs is clarified. In particular, we find that the latter approach corresponds to considering a stochastic process (in real time) which evolves towards an equilibrium state governed by the topological charge which precisely corresponds to the starting classical action in the Baulieu-Singer approach.

The plan of the paper is as follows. In Sec. 2 we describe how TQFTs can be constructed from Bogomol'nyi equations. We then show in Sec. 3 that Bogomol'nyi equations can be interpreted as Langevin equations. The fact that the potential in the latter equations is related to the topological charge allows us to connect this approach with that of Baulieu-Singer.² Finally, in Sec. 4 we give a summary of our work and make some concluding remarks.

* Becario de Perfeccionamiento Comisión de Investigaciones Científicas de Buenos Aires.

† Investigador Comisión de Investigaciones Científicas de Buenos Aires.

2. Bogomol'nyi Equations and TQFTs

Bogomol'nyi equations are first order differential equations originally derived⁹ from an inequality relating the classical action S and the topological charge Q_T . One can prove that any solution to these equations also solves the (more involved) second order Euler-Lagrange equations for S .

By following this route, exact vortex and monopole solutions can be found for gauge theories with spontaneous symmetry breaking⁹⁻¹² and instanton solutions for pure Yang-Mills and CP^N models.^{13,14}

Schematically, Bogomol'nyi equations can be obtained as follows. Given an action of the form

$$S = \int_M \sum_i (a^i)^2, \quad (2.1)$$

with a^i ($i = 1, 2, \dots$) some functionals of the fields that describe the system, defined on a d -dimensional space-time manifold M , one rewrites S as

$$S = \frac{1}{2} \int_M \sum_i (a^i \pm \tilde{a}^i)^2 \mp \int_M \sum_i \tilde{a}^i a^i, \quad (2.2)$$

with $\tilde{a}^i$ some functionals conveniently chosen.

The main point now is that under certain conditions the second term on the r.h.s. of Eq. (2.2) can be identified with a topological invariant Q_T ,

$$Q_T = \alpha \int_M \sum_i \tilde{a}^i a^i. \quad (2.3)$$

(Here α is a constant to be chosen according to the model.) One then has, from Eq. (2.2),

$$S \geq \mp Q_T. \quad (2.4)$$

The bound is saturated whenever the Bogomol'nyi equations

$$(a^\pm)^i \equiv a^i \pm \tilde{a}^i = 0 \quad i = 1, 2, \dots \quad (2.5)$$

hold.

As an example, consider a gauge field A taking values on the Lie algebra of some group G , in $d = 4$ Euclidean space. Take $a = \frac{1}{2}F$ and $\tilde{a} = \frac{1}{2}*F$, the Hodge dual of the curvature F , and $\alpha^{-1} = -16\pi^2$. Then Eq. (2.5) becomes the instanton defining equation¹³ and Q_T the Chern-Pontryagin invariant. Consider instead a $U(1)$ gauge field A and a complex scalar ϕ in $d = 2$ Euclidean space. Choose $a^1 = F/\sqrt{2}$, $a^2 = D_\mu \epsilon_{ab} \phi_a / \sqrt{2}$, with F the curvature and D the covariant derivative. Take $\tilde{a}^1 = *(\eta^2 - \phi^2)(e/\sqrt{2})$ and $\tilde{a}^2 = *(D\phi_a)\sqrt{2}$, where $*$ stands for the Hodge star

operator, e is the gauge coupling constant and η is the vacuum expectation value of the Higgs fields. Then Eq. (2.5) becomes the Bogomol'nyi equations for Abelian Higgs vortices⁹ (in this case $\alpha^{-1} = 2\pi\eta^2$). The same can be done for non-Abelian vortices, Prasad-Sommerfield monopoles,^{9,10} and CP^n instantons.¹⁴

As we stated above, TQFTs can be obtained as a result of BRST quantization of a "trivial" Gaussian action.^{3,4} It is through this Gaussian action that Bogomol'nyi equations come into play. The strategy is as follows:

(i) Given a collection of fields Φ that describe the system, one considers the Bogomol'nyi equations (of the form (2.6)) associated with certain action S in the way described above.

(ii) Auxiliary fields G^i are introduced by constructing the Gaussian action S_G ,

$$S_G = \int_M [G^i - (a^\pm)^i]^2 . \quad (2.6)$$

Since $(a^\pm)^i$ stem as self-dual quantities (see below) G^i are taken as self-dual fields. Consequently, S_G is independent of the metric. Equations of motion for G^i read

$$(a^\pm)^i - G^i = 0 . \quad (2.7)$$

(iii) It is easy to see that S_G is invariant under the transformations:

$$\delta\Phi = \delta\lambda \quad (2.8a)$$

$$\delta G^i = \frac{\partial(a^\pm)^i}{\partial\Phi} \delta\lambda . \quad (2.8b)$$

Transformations (2.8a) represent the largest local symmetry possible for the fields describing the system. Transformations (2.8b) are accommodated so that $\delta S_G = 0$. One then carries out BRST quantization of S_G with the invariances (2.8). Note that transformations (2.8b) are enough to gauge away auxiliary fields G . Equation (2.7) becomes, in the $G = 0$ gauge, Bogomol'nyi equations.

(iv) After BRST quantization, the generating functional Z_{TOP} defining the TQFT reads

$$Z_T = \int D\text{fields} \exp(-S_{\text{eff}}) , \quad (2.9)$$

where the integration measure includes original fields Φ as well as auxiliary and ghost fields. The effective action reads

$$S_{\text{eff}} = S_G + \{Q, V\} . \quad (2.10)$$

Here $\{Q, \}$ represents the BRST transformation of some functional of the fields.

It is important to note at this point that:

(a) The shift $G^i \rightarrow G^i \pm (a^\pm)^i$ reduces S_G to a trivial action; this shows that all interesting effects should arise from quantization.

(b) Although BRST quantization may introduce a metric dependence in S_{eff} , Z_{TOP} is indeed metric-independent since¹

$$\frac{\delta Z_{\text{TOP}}}{\delta g_{\mu\nu}} = \left\langle \left\{ Q, \frac{1}{\sqrt{g}} \frac{\delta V}{\delta g_{\mu\nu}} \right\} \right\rangle = 0. \quad (2.11)$$

We have used here the fact that the v.e.v. of any BRST variation is zero. It is in this sense that one says that Z_{TOP} defines a TQFT.

3. Stochastic Processes and TQFTs

In Ref. 4 a connection between S_G , the Gaussian action used in Ref. 3 to derive Yang-Mills TQFT, and a Langevin equation with G acting as a random field was signalled. We shall now discuss this connection, trying to clarify its origin and relate it to the Baulieu-Singer approach to TQFTs.

Consider the Langevin equation for some generic field Φ ,

$$\frac{\partial \Phi}{\partial t} - \frac{\delta W}{\delta \Phi} = G, \quad (3.1)$$

with t the real time, G a random field and W a “potential” providing a drift to the stochastic process to be described by Eq. (3.1).¹⁵

How does one have to choose W in order to connect this Langevin equation with TQFTs?

Let us begin by discussing an example. Given a Yang-Mills gauge field A in $d = 4$ (compactified) space, the Chern-Pontryagin invariant Q_T reads (cf. Eq. (2.3))

$$Q_T = \frac{1}{4\pi^2} \int a \tilde{a} = \alpha \int d^4x {}^*F_{\mu\nu} F_{\mu\nu}, \quad (3.2)$$

where $\alpha = -(16\pi^2)^{-1}$.

Now Q_T can be written in terms of the Chern-Simons secondary characteristic class K_μ ¹⁶:

$$K_\mu = \frac{-1}{16\pi^2} \text{tr} \, \varepsilon_{\mu\nu\rho\sigma} \left(F_{\nu\rho} A_\sigma - \frac{2}{3} A_\nu A_\rho A_\sigma \right), \quad (3.3)$$

in the form

$$Q_T = \int d^4x \partial_\mu K_\mu. \quad (3.4)$$

Take the 0-component of K_μ in the $A_o = 0$ gauge. One can easily prove that¹⁶

$$\lim_{t \rightarrow \infty} W(t) = \frac{1}{\alpha} Q_T, \quad (3.5)$$

with

$$W(t) = \frac{1}{\alpha} \int d^3x K_0. \quad (3.6)$$

(We have chosen initial conditions $A_i = 0$ in such a way that $W(-\infty) = 0$.)

If one now chooses $W(t)$ to be the “potential” W in the Langevin equation (3.1) for Yang-Mills field A one writes

$$\frac{\partial A_i}{\partial t} - \frac{\delta W(t)}{\delta A_i} - G_{oi} = 0. \quad (3.7)$$

One discovers, using Eqs. (3.3)–(3.6), that Langevin equation (3.7) coincides with the instanton defining equation in the presence of a noise:

$$G_{\mu\nu} - (F_{\mu\nu} - {}^*F_{\mu\nu})|_{A_o=0} = 0. \quad (3.8)$$

In view of the behavior (3.5), one can conclude that the stochastic process described by Eq. (3.7) evolves towards an equilibrium state governed by $\exp(-Q_T)$.

We are now able to understand the connection between Baulieu-Singer² BRST quantization of the topological action $S_T = Q_T$ and Labastida-Pernici³ quantization of a Gaussian action S_G , two alternative ways of getting TQFTs. Indeed, it is known from Parisi-Sourlas’ work¹⁵ that one can arrive at supersymmetric quantum field theories by starting from a real time Langevin equation of the kind (3.1) for the bosonic sector. The corresponding stochastic generating functional Z_{STOCH} (defined as a path integral over bosonic and random fields, with an action which is the squared Langevin equation) has a large symmetry to be fixed: an arbitrary change in bosonic fields is always compensated for by an appropriate change in the random fields. In the gauge fixing procedure ghosts (and ghosts of ghosts in certain cases) come into play so that the resulting action is supersymmetric.

For the example we are dealing with, the Parisi-Sourlas approach corresponds to taking the square of the l.h.s. of Langevin equation (3.7) as starting (Gaussian) action and then proceeding to BRST quantization. Z_{STOCH} then reads

$$Z_{\text{STOCH}} = \int D\text{fields} \exp \left\{ - \int d^4x \left(G_i - \frac{\partial A_i}{\partial t} - \frac{\delta W(t)}{\delta A_i} \right)^2 + \text{BRST terms} \right\}. \quad (3.9)$$

Now, in view of (3.8), this is nothing but the generating functional Z_{TOP} (in the $A_o = 0$ gauge) defined by Eq. (2.9) and obtained following the Labastida-Pernici approach to TQFTs.

From the stochastic viewpoint, one can then interpret (3.9) as arising from a stochastic process evolving to an equilibrium state governed by $W(t = \infty) = Q_T$. But precisely Q_T was the starting action considered by Baulieu-Singer² in their approach to TQFTs.

Interestingly enough, this link can be established for other TQFTs. Indeed, consider a gauge field A taking values in the Lie algebra of $\text{SO}(3)$ and a Higgs field ϕ in the adjoint representation. In three-dimensional space a topological invariant (which can be interpreted as the magnetic charge) can be defined as follows:

$$Q_T = -\alpha \operatorname{tr} \int d^3x \varepsilon_{\mu\nu\rho} F_{\mu\nu} D_\rho \phi \quad \rho = 1, 2, 3 \quad (3.10)$$

(in this case $\alpha = e/8\pi\eta$).

Again, Q_T can be written in terms of a vector K_μ ,

$$K_\mu = -\alpha \operatorname{tr} \varepsilon_{\mu\nu\rho} F_{\nu\rho} \phi. \quad (3.11)$$

Following the recipe described above, we now construct Langevin equations with potential $W(t)$,

$$W(t) = \frac{1}{\alpha} \int dx^2 K_o, \quad (3.12)$$

and $\lim_{t \rightarrow \infty} W(t) = \frac{1}{\alpha} Q_T$. They read

$$\frac{\partial \phi^\alpha}{\partial t} - \frac{1}{2} \varepsilon_{ij} F_{ij}^\alpha - G^\alpha = 0 \quad (3.13a)$$

$$\frac{\partial A_i^\alpha}{\partial t} - \varepsilon_{ij} D_j \phi^\alpha - G_i^\alpha = 0. \quad (3.13b)$$

As expected, the l.h.s. in Eqs. (3.13) coincides with the Bogomol'nyi equations⁹ for the Prasad-Sommerfield monopole¹⁰ in the $A_o = 0$ gauge. Starting with a topological action $S_T = Q_T$, with Q_T given by (3.10), Baulieu-Grossman have constructed a TQFT which can also be obtained, as shown in Ref. 4, by gauge-fixing a Gaussian action constructed from the square of Langevin equations (3.13). From the discussion above, it is clear now why the two approaches lead to the same TQFT.

The descent can be performed once more in order to derive two-dimensional TQFTs. Consider a $\text{U}(1)$ gauge field A and charge scalar ϕ^α ($\alpha = 1, 2$) in two dimensions. A topological invariant (which can be interpreted as the magnetic vortex flux) can be defined:

$$Q_T = \alpha \int d^2x \varepsilon_{\mu\nu} \left\{ \frac{e}{2} F_{\mu\nu} (\eta^2 - \phi^2) - D_\mu \phi^a D_\nu \phi^b \varepsilon_{ab} \right\}, \quad (3.14)$$

where $\alpha = 1/2\pi\eta^2$.

Again Q_T can be written in terms of a current K_μ :

$$Q_T = \int d^2x \partial_\mu K^\mu \quad (3.15)$$

$$K_\mu = \frac{\alpha}{2} \varepsilon_{\mu\nu} (e\eta^2 A_\nu + \varepsilon_{ab} \phi^a D_\nu \phi^b). \quad (3.16)$$

Taking $W = W(t)$,

$$W(t) = \frac{1}{\alpha} \int dx K_0, \quad \lim_{t \rightarrow \infty} W(t) = \frac{1}{\alpha} Q_T, \quad (3.17)$$

the corresponding Langevin equations read

$$\frac{\partial \phi_a}{\partial t} - \varepsilon_{ab} (D_1 \phi)^b = G_a \quad (3.18a)$$

$$\frac{\partial A_1}{\partial t} - \frac{e}{2} (\eta^2 - \phi^2) = G_1, \quad (3.18b)$$

which coincide (in the $A_0 = 0$ gauge and for $G = 0$, $G_a = 0$) with Bogomol'nyi equations for Abelian vortices.⁹ In Ref. 18 G. Thompson and one of us have obtained a TQFT associated with two-dimensional instantons (vortices) just by constructing Z_{TOP} (Eq. (3.9) or equivalently Eq. (2.9)) from Langevin equations (3.8) (see also Ref. 19). The alternative Baulieu-Singer approach for this model has been discussed in Ref. 20, leading to an equivalent TQFT.

The extension to non-Abelian vortices is straightforward.^{20–22} Also CP^n instanton equations can be considered in this way and the corresponding TQFT constructed from Z_{STOCH} .

At this point it is important to stress that the TQFT associated with a given classical action S_{cl} (Yang-Mills action, Abelian Higgs model action, etc.) is not obtained by inserting S_{cl} in the Langevin equation. Instead, one has to construct a potential $W = W(t)$ such that $W(t \rightarrow \infty) = Q_T$, the relevant topological charge. With this potential the Langevin equations are nothing but the Bogomol'nyi equations associated with S_{cl} . In those cases for which a critical relation between coupling constants must hold in order that Bogomol'nyi equations exist ($\lambda = e^2$ for vortices, $\lambda = 0$ for monopoles), the effective action S_{eff} resulting from BRST gauge fixing of the squared Langevin equation is defined at the critical point.

In general S_{eff} can be written in terms of S_{cl} and ghost terms in the following form:

$$S_{\text{eff}} = S_{\text{cl}} + Q_T + \text{ghost terms} + \text{gauge fixing terms} \quad (3.19)$$

and can be expressed in the simple form

$$S_{\text{eff}} = \{Q, F\}, \quad (3.20)$$

with F some functional of the original fields, ghost fields and auxiliary fields.

4. Summary and Conclusions

Let us conclude with some observations. The main feature of TQFTs is that they have the largest local symmetry possible for the fields that describe the system. This symmetry is responsible for gauging away any dependence on local symmetries.²³ The observables of the resulting quantum theory are then topological invariants computable as expectation values mainly of the ghost fields. From the stochastic point of view the large symmetry is present due to the fact that one can accommodate random field variations so as to compensate for arbitrary (local) variations of the original fields. In the examples discussed by Parisi-Sourlas,¹⁵ the symmetry was an ordinary gauge symmetry so that the Faddeev-Popov method could be applied, leading to a quadratic ghost Lagrangian. If one undoes the Parisi-Sourlas derivation by integrating out ghosts one ends with a “trivial” Gaussian generating functional.

In the case of the TQFTs that we have discussed in the present work, one discovers a second generation gauge invariance,²⁴ which for random fields is valid on-shell. One cannot then apply the Faddeev-Popov method naively. Instead, one may use the Batalin-Vilkovisky²⁴ machinery specialized to the case of a “first stage reducible theory” (which corresponds, in their classification, to the presence of the second generation gauge invariance just mentioned). The resulting effective Lagrangian then contains complicated and, in general, non-quadratic terms with ghosts, ghosts of ghosts, auxiliary fields, etc. This prevents one in principle from integrating out ghost fields in order to undo Parisi-Sourlas derivation and get into a simple Gaussian stochastic generating functional. We think it is interesting to study more thoroughly this issue in order to understand what is happening from the stochastic point of view.

Another point concerns Nicolai mappings. These are mappings which transform the bosonic part of a supersymmetric Lagrangian (not undergoing dynamical supersymmetry breaking) into Gaussian ones. Moreover, the associated Jacobian coincides with the determinant arising from integration of the fermionic sector. As explained in Refs. 15 and 17 Nicolai mappings are intimately related to stochastic processes. In fact, the Nicolai mapping can be interpreted as a stochastic equation, not just formally but in the sense that it gives the correct Langevin equation providing the stochastic equation for the original Lagrangian. This is the clue that allowed Birmingham, Rakowski and Thompson⁴ to construct Nicolai mappings from TQFTs. Now we have seen that the Langevin equation is nothing but the Bogomol’nyi equation for the models discussed here. We can then understand why the critical values for coupling constants required in order to have Bogomol’nyi equations (in particular $\lambda = e^2$ both for Abelian and non-Abelian vortices)

coincide with those required by supersymmetry.^{25,26} This connection is due to the fact that Bogomol'nyi equations forced by a noise are Nicolai maps characterizing the supersymmetric theory. Let us finally point out that at these critical values the (classical) energy-momentum tensor $T_{\mu\nu}$, associated with Bogomol'nyi equations, vanishes.¹¹ It is then natural to see a TQFT arise after quantization, since TQFTs are characterized by a metric-independent partition functional (and hence a vanishing energy-momentum tensor v.e.v.).

In summary we have been able to connect different approaches to TQFTs by exploiting the relation between Bogomol'nyi equations and Langevin equations for stochastic processes involving the relevant topological charge. In view of the large symmetry characterizing TQFTs, the ghost Lagrangian is highly nontrivial and this is why the original Parisi-Sourlas¹⁵ derivation of Nicolai mappings (in theories for which the gauge fixing just amounts to including a Faddeev-Popov determinant) is here somehow obscured. We think that this point, as well as that concerning more general Bogomol'nyi equations and the associated TQFTs, deserves more investigation.

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